

Feasibility of Remote Pulse Transit Time Estimation Using Narrow-band Multi-wavelength Camera Photoplethysmography

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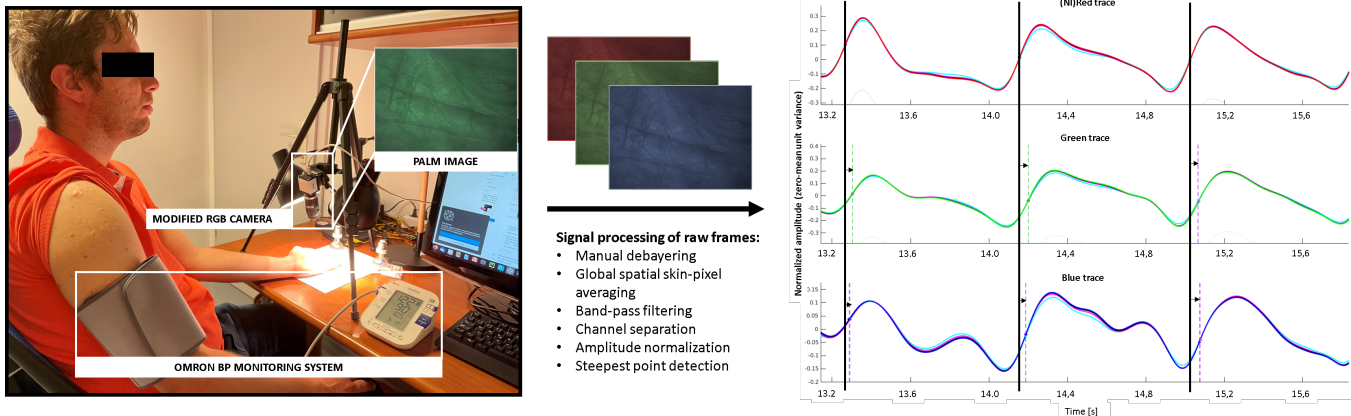


Figure 1: Our proposed multi-wavelength remote pulse transit time (PTT) monitoring system, and resulting signal waveforms from each wavelength. Differences between the waveforms (solid and dashed vertical lines) correspond to PTT.

ABSTRACT

Contact-free remote sensing is showing increasing potential for real-life use, especially for screening applications (e.g., thermal cameras to detect fever). We investigated the feasibility of estimating pulse transit time (PTT) – a marker known to be highly correlated with blood pressure – in a remote way from a single measuring site, using just a single modified RGB camera. We replaced the default infrared (IR) filter with a narrow-band triple-bandpass filter allowing us to remotely measure the PTT between green and near-infrared (NIR) wavelengths. Such a filter minimizes the inter-channel influence and band overlap, while also leveraging the non-traditional NIR information. We based the PTT measurements on the fact that different wavelengths penetrate to different skin layers. This allowed for the capture of slightly delayed blood pulses corresponding to different wavelengths and skins depth. Our early observations in 5-subject small-scale experiment show relatively consistent temporal delays between the waveforms from different wavelengths in accordance with expectations and related work.

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These early results show promise for further research in remote multi-wavelength PTT and blood pressure (BP) estimation, while also highlighting important fundamental and technical challenges to be considered.

CCS CONCEPTS

• **Applied computing** → Health informatics; • **Health informatics**;

KEYWORDS

Remote sensing; Pulse Transit Time; Multi-wavelength; Photoplethysmography

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1 INTRODUCTION

Contact-free physiological monitoring has seen rapid growth over the past decade [3], starting from heart rate (HR) monitoring and recently expanding towards more detailed vitals, such as heart rate variability, oxygen saturation and blood pressure (BP). These applications are mainly facilitated by remote photoplethysmography

(rPPG), which reflects cardiovascular activity and blood perfusion via changes in skin color [1].

Our aim was to investigate the feasibility of remote multi-wavelength (MW) pulse transit time (PTT) estimation. The latter is known to be highly correlated with BP [4]. Instead of using traditional multi-site measuring technique [5], we propose and investigate a single-site MW approach in which we measure the PTT between different skin depths instead of different spatial locations. Such an approach would be valuable, as currently the “golden standard” for in-office and at-home BP measurement remains the use of a cuff-based sphygmomanometer. Our proposed approach could allow for unobtrusive monitoring, be more comfortable and have less influence on the measurement result (e.g., white coat syndrome), using just a single RGB camera.

The idea behind PTT is to measure the time it takes for a specific heart pulse wave to traverse from its source to a distal location. Typically this is done using two sensors, most commonly ECG and a distal PPG, and then measure the distance between the R peak in the ECG signal and the corresponding systolic peak in the synchronized PPG signal.

2 RECORDING SETUP AND DATA COLLECTION

Since skin layers are thin and blood traversal velocity high, the PTT we attempted to observe is short. We thus required a camera with a very high frame rate and as little on-board processing as possible. We used the iDS 3040SE RGB camera, which allowed us to record at 250 frames per second (fps) at 540x720 resolution. As we are interested in the near-infrared (NIR) range as well – corresponding to deeper skin layers – we removed the default IR filter on the image sensor and replaced it with a triple-bandpass MidOpt TB475/550/850 filter, which allows only light in narrow bands of 475 ± 10 nm (blue), 550 ± 10 nm (green) and 850 ± 22 nm (NIR) to pass. It is also vital to use a full-spectrum light source, as modern LEDs do not emit NIR wavelengths. We thus used incandescent Osram Decostar 51ALU bulbs. These were plugged into a DC power source to avoid flickering at 50 Hz caused by AC power, which is picked up as noise by the high-fps camera.

We used the described recording setup to record palm skin of test subjects who gave written consent for recordings and data processing. The palm is known to exhibit the most pulsatility (similar to fingertip and forehead) and is relatively comfortable to record. We collected video recordings alongside their corresponding ground-truth BP measurements taken with a clinical-grade Omron digital sphygmomanometer. The recordings lasted for 30 s each, during which a single ground-truth measurement was taken, since the measurement itself takes roughly 20 s. Thus a single systolic BP (SBP), diastolic BP (DBP) and HR ground truth were applied to the whole 30-sec recording. Each subject was recorded several times in two different scenarios: **A) Resting position** – subject was seated upright with a cuff in a comfortable position and relaxed for 1 minute. This allowed for capture of stabilized baseline rest vitals. **B) Intense physical activity** – subject performed an intense sequence of jump squats and jumping jacks without rest. This elevated their vitals substantially. The recording was made immediately after completion, as the vitals do not drop that rapidly. Using

this protocol we so far collected 90 high-quality 30-sec recordings of 5 subjects, split between the two scenarios.

3 SIGNAL PROCESSING AND RESULTS

Once the data was collected, we ran it through a signal processing pipeline. The details of this pipeline are subject of ongoing development and optimization and will thus not be described in great detail, however, the major steps include the following: 1.) manual debayering of frames, 2.) global spatial averaging of all (skin) pixels, 3.) zero-phase band-pass filtering in the range [0.5, 6.0] Hz, 4.) channel separation to avoid inter-channel influence, 5.) amplitude normalization to the [-1, 1] range, and 6.) detection of the reference steepest point on the systolic rise with a derivative method.

Our initial hypothesis based on related work [2] and physiological background was that the cardiac wave from the deeper vessels in the NIR range should be observed earlier, while those in the shallower layers should be delayed. If we define these times as $PTT = timestamp_steepest_{green} - timestamp_steepest_{NIR}$, we expect these times to be positive if in agreement with our hypothesis. Our early observations on the data collected so far confirm this, as shown in Figure 2. Furthermore, we also observe a distinct difference in PTTs between different scenarios, as those measured post-activity with elevated vitals are shorter due to elevated BP and increased vascular stiffness, as expected.

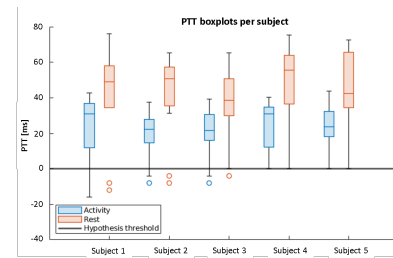


Figure 2: Per-subject boxplots of PTTs between NIR and green, taking the steepest systolic points as reference for the measurement.

4 CONCLUSION

The aim of this work was to investigate the feasibility of narrow-band remote MW PTT estimation. Our initial results indicate agreement with the hypothesis that PTT between skin layers can be measured and appears to be correlated to BP, but detailed analysis is ongoing. We found that using narrow wavelength bands helps separate the channels better to more easily observe PTT, and using the NIR part of the spectrum (instead of traditional red) further increases the PTT between red and green, which is fundamentally short. We are currently working on confirming correlation with BP variations (normal, elevated), which could indicate real-world application potential of such a system.

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